Atom-adic eigenvalue valuations for the staircase operator on $V_{(4,2)}$ at $n = 6$

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Abstract

Let $\Pi_q \in H_q(\mathfrak{S}_6)$ be the staircase Bott–Samelson product $\prod_{j=1}^{15}(1+T_{i_j})$ along $\underline{w}_0 = 1\,21\,321\,4321\,54321$, and let $\mathfrak{a}(q) = q^2+4q+1 \in \mathbb{Z}[q]$. Acting on the Specht module $V_{(4,2)}$, the operator Π_q has rank 3. We prove that, in any extension of $\mathbb{Q}(q)$ where the characteristic polynomial of $\Pi_q|_{V_{(4,2)}}$ splits, the three non-zero eigenvalues e_1, e_2, e_3 satisfy

$$\nu_{\mathrm{atom}}(e_1) = 2, \qquad \nu_{\mathrm{atom}}(e_2) = 1, \qquad \nu_{\mathrm{atom}}(e_3) = 0,$$

where ν_{atom} is the $\mathfrak{a}$-adic valuation extended to the algebraic closure. The proof has two ingredients: (i) a symbolic computation, via Lagrange interpolation over $\mathbb{Q}$, of the elementary symmetric functions $\sigma_1, \sigma_2, \sigma_3$ of the eigenvalues of $\Pi_q|_{V_{(4,2)}}$, with their explicit factorisations; and (ii) the Newton polygon theorem applied to the rank-3 characteristic polynomial.

This refines the empirical fact $\sigma_3(\Pi_q|_{V_{(4,2)}}) = q^9(1+q)^{21}\mathfrak{a}(q)^3$ into an eigenvalue-level statement: the cube exponent comes from the additive split $3 = 2+1+0$ of three eigenvalue valuations, not from coincidence at the determinant.

1 Setup

Let $H_q = H_q(\mathfrak{S}_n)$ be the Hecke algebra of the symmetric group with parameter q, with generators $T_1, \ldots, T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid and commutation relations. For $n = 6$ fix the staircase reduced word

$$\underline{w}_0 = (i_1, \ldots, i_{15}) = (1,\ \ 2,1,\ \ 3,2,1,\ \ 4,3,2,1,\ \ 5,4,3,2,1)$$

for the longest element $w_0 \in \mathfrak{S}_6$, and let

$$\Pi_q \ = \ (1+T_{i_1})(1+T_{i_2})\cdots(1+T_{i_{15}}) \ \in \ H_q.$$

For each partition $\lambda \vdash n$, write V_λ for the cell (Specht) module of dimension f^λ. We work throughout in Hoefsmit's seminormal form, where T_i acts by an explicit matrix over $\mathbb{Q}(q)$ in the basis indexed by standard Young tableaux [1].

We focus on $\lambda = (4,2)$ and $n = 6$. Then $f^{(4,2)} = 9$, so $\Pi_q|_{V_{(4,2)}}$ is a 9×9 matrix with entries in $\mathbb{Q}(q)$.

Let $\mathfrak{a} = \mathfrak{a}(q) := q^2+4q+1 \in \mathbb{Z}[q]$. This polynomial is irreducible over $\mathbb{Q}$ (its roots are $-2 \pm \sqrt{3}$, irrational) and primitive. Hence the ideal $(\mathfrak{a})$ is a prime ideal of height 1 in the principal ideal domain $\mathbb{Q}[q]$, and the localisation $\mathbb{Q}[q]_{(\mathfrak{a})}$ is a discrete valuation ring with uniformiser $\mathfrak{a}$. We write

$$\nu_{\mathrm{atom}} \colon \mathbb{Q}(q)^\times \longrightarrow \mathbb{Z}$$

for the resulting $\mathfrak{a}$-adic valuation: for a non-zero polynomial $f \in \mathbb{Q}[q]$, $\nu_{\mathrm{atom}}(f)$ is the largest integer m with $\mathfrak{a}^m \mid f$, and ν_{atom} extends uniquely to $\mathbb{Q}(q)$ by $\nu_{\mathrm{atom}}(f/g) = \nu_{\mathrm{atom}}(f) - \nu_{\mathrm{atom}}(g)$.

Definition 1. For $k \geq 1$ define $\sigma_k(\Pi_q|_{V_\lambda}) \in \mathbb{Q}(q)$ as the k-th elementary symmetric polynomial of the eigenvalues of $\Pi_q|_{V_\lambda}$ (counted with multiplicity in any algebraic-closure extension), equivalently the trace of the k-th exterior power $\Lambda^k(\Pi_q|_{V_\lambda})$. These are computed concretely by Newton's identities from $p_i := \mathrm{tr}((\Pi_q|_{V_\lambda})^i)$:

$$\sigma_k = \frac{1}{k}\sum_{i=1}^{k}(-1)^{i-1}\sigma_{k-i}\, p_i, \qquad \sigma_0 = 1.$$

Because the entries of $\Pi_q|_{V_\lambda}$ in Hoefsmit form lie in $\mathbb{Q}(q)$ with denominators that are products of $1 - q^{-d}$ (for axial-distance integers d), clearing denominators shows $\sigma_k(\Pi_q|_{V_\lambda}) \in \mathbb{Z}[q, q^{-1}]$; in fact multiplying out we find $\sigma_k \in \mathbb{Z}[q]$, which is also a consequence of the fact that Π_q acts on the integral form of V_λ.

2 Main theorem

Theorem 2 (T1)**.** *Let $g(t) \in \mathbb{Q}(q)[t]$ be the minimal monic polynomial vanishing on the non-zero eigenvalues of $\Pi_q|_{V_{(4,2)/n=6}}$, equivalently the degree-3 factor of the 9×9 characteristic polynomial after stripping the kernel factor t^6. Extend the $\mathfrak{a}$-adic valuation ν_{atom} to the algebraic closure of $\mathbb{Q}(q)$. Then $g(t)$ is monic of degree 3, and its three roots e_1, e_2, e_3 satisfy*

$$\nu_{\mathrm{atom}}(e_1) = 2, \qquad \nu_{\mathrm{atom}}(e_2) = 1, \qquad \nu_{\mathrm{atom}}(e_3) = 0.$$

Equivalently,

$$g(t) \;=\; (t - \mathfrak{a}^2\alpha)(t - \mathfrak{a}\,\beta)(t - \gamma)$$

in the algebraic closure, with α, β, γ all of $\mathfrak{a}$-valuation 0.

The proof has two parts. First, a symbolic computation pins down $\sigma_1, \sigma_2, \sigma_3$ and shows that $\sigma_4 = 0$ identically (so $\mathrm{rk}(\Pi_q|_{V_{(4,2)}}) = 3$). Second, a Newton polygon argument forces the multiset of root valuations.

3 Symbolic computation: σ_k for $k = 1, \ldots, 4$

Lemma 3. *The elementary symmetric polynomials $\sigma_k = \sigma_k(\Pi_q|_{V_{(4,2)/n=6}})$ satisfy, as polynomials in q:*

$$\sigma_1 \;=\; q^2(1+q)^3 \cdot P_1(q), \tag{1}$$
$$\sigma_2 \;=\; q^5(1+q)^{10} \cdot \mathfrak{a}(q) \cdot P_2(q), \tag{2}$$
$$\sigma_3 \;=\; q^9(1+q)^{21} \cdot \mathfrak{a}(q)^3, \tag{3}$$
$$\sigma_4 \;=\; 0, \tag{4}$$

where $P_1(q)$ and $P_2(q)$ are explicit palindromic polynomials of degree 8, both coprime to $\mathfrak{a}(q)$:

$$P_1(q) \;=\; 1 + 13q + 79q^2 + 245q^3 + 365q^4 + 245q^5 + 79q^6 + 13q^7 + q^8,$$

$$P_2(q) \;=\; 1 + 17q + 103q^2 + 313q^3 + 473q^4 + 313q^5 + 103q^6 + 17q^7 + q^8.$$

Consequently

$$\nu_{\mathrm{atom}}(\sigma_1) = 0, \qquad \nu_{\mathrm{atom}}(\sigma_2) = 1, \qquad \nu_{\mathrm{atom}}(\sigma_3) = 3.$$

Proof. The proof is computational, but rigorous in the strongest sense available: it is exact rational arithmetic, not numerics.

Let $P^{(\nu)} \in \mathrm{Mat}_9(\mathbb{Q})$ denote the matrix obtained from $\Pi_q|_{V_{(4,2)/n=6}}$ by specialising q to a positive rational integer $\nu \geq 2$. The Hoefsmit denominators $1 - q^{-d}$ for axial-distance integers $d \in \{1, 2, 3, 4, 5\}$ do not vanish at any integer $\nu \geq 2$, so $P^{(\nu)}$ is well-defined as a matrix in $\mathbb{Q}$.

The map $\nu \mapsto \sigma_k(P^{(\nu)})$ takes integer values at $\nu = 2, 3, 4, \ldots$ that agree with the polynomial $\sigma_k(q) \in \mathbb{Z}[q]$ evaluated at $q = \nu$. The polynomial $\sigma_k(q)$ has q-degree bounded by $k \cdot 15$ (each of the 15 factors $1 + T_{i_j}$ has matrix entries of q-degree at most 1, so entries of Π_q have q-degree at most 15, and σ_k as a sum of principal $k \times k$ minors has q-degree at most $15k$). In particular for $k \leq 4$ the degree is at most 60.

We evaluate $\sigma_k(P^{(\nu)})$ at $\nu = 2, 3, \ldots, 111$ (110 distinct integer points), using exact rational arithmetic in SymPy, and apply Lagrange interpolation over $\mathbb{Q}$ to recover the unique polynomial $\sigma_k(q) \in \mathbb{Q}[q]$ of degree ≤ 60 that agrees with the evaluations. Because the polynomial is uniquely determined by any 61 distinct evaluations and we use 110, the interpolation is overdetermined and consistent — this provides a built-in cross-check. The recovered polynomials are:

- $\sigma_1(q)$ has degree 13 and factors as $q^2(1+q)^3 P_1(q)$, with P_1 the explicit palindrome above. Substituting the root $q = -2 + \sqrt{3}$ of $\mathfrak{a}$ into P_1 gives a non-zero algebraic number (verified symbolically in SymPy by exact arithmetic), proving $\mathfrak{a} \nmid P_1$.
- $\sigma_2(q)$ has degree 25 and factors as $q^5(1+q)^{10}\mathfrak{a}(q)P_2(q)$, with P_2 the explicit palindrome above. Substituting $q = -2 + \sqrt{3}$ into P_2 gives a non-zero algebraic number (verified), so $\mathfrak{a} \nmid P_2$.
- $\sigma_3(q)$ has degree 36 and factors as $q^9(1+q)^{21}\mathfrak{a}(q)^3$, with no remaining factor.
- $\sigma_4(q)$ is identically zero: each of the 110 integer evaluations of σ_4 returns $0 \in \mathbb{Q}$ exactly. Since $\sigma_4(q)$ is a polynomial of degree at most 60, its vanishing at 61 or more distinct points forces $\sigma_4 \equiv 0$.

Combining, $\nu_{\mathrm{atom}}(\sigma_1) = 0$, $\nu_{\mathrm{atom}}(\sigma_2) = 1$, $\nu_{\mathrm{atom}}(\sigma_3) = 3$. □

Remark 4. Lemma 3 is also consistent with the data table in `2026-04-27-joint-divisibility/RESULT.md` for $V_{(4,2)/n=6}$, which records the same factorisations. The script `2026-04-27-newton-polygon-proof/verify.py` reruns the calculation and adds the $\sigma_4 = 0$ verification.

Corollary 5. $\mathrm{rk}(\Pi_q|_{V_{(4,2)/n=6}}) = 3$.

Proof. For any matrix A over $\mathbb{Q}(q)$ of size d, $\sigma_k(A) = \mathrm{tr}(\Lambda^k A) = 0$ for all $k > \mathrm{rk}(A)$, while $\sigma_{\mathrm{rk}(A)}(A) \neq 0$. By Lemma 3, $\sigma_3 = q^9(1+q)^{21}\mathfrak{a}^3 \neq 0$ and $\sigma_4 = 0$, so $\mathrm{rk}(\Pi_q|_{V_{(4,2)/n=6}}) = 3$. □

Corollary 6. *The characteristic polynomial of* $\Pi_q|_{V_{(4,2)/n=6}}$ *in* $\mathbb{Q}(q)[t]$ *has the form*

$$\chi(t) \;=\; t^6 \cdot g(t), \qquad g(t) \;=\; t^3 - \sigma_1\, t^2 + \sigma_2\, t - \sigma_3,$$

with $\sigma_1, \sigma_2, \sigma_3$ *as in Lemma 3. In particular,* $g(t) \in \mathbb{Q}(q)[t]$ *is monic of degree* 3.

Proof. The characteristic polynomial of any 9×9 matrix A has the form $\chi_A(t) = \sum_{k=0}^{9} (-1)^k \sigma_k(A)\, t^{9-k}$, where $\sigma_0 = 1$. By Corollary 5 and Lemma 3, $\sigma_k = 0$ for $k \geq 4$, so

$$\chi(t) = t^9 - \sigma_1 t^8 + \sigma_2 t^7 - \sigma_3 t^6 = t^6(t^3 - \sigma_1 t^2 + \sigma_2 t - \sigma_3). \qquad \square$$

4 Newton polygon

We recall the Newton polygon theorem in the form we need. Let K be a field with a discrete valuation $v\colon K^\times \to \mathbb{Z}$, let $\bar{K}$ be an algebraic closure, and extend v to $\bar{K}^\times \to \mathbb{Q}$ (such an extension exists, and the multiset of values $v(e_i)$ for the roots e_i of any given polynomial $f \in K[t]$ is independent of the choice of extension; cf. [2, Ch. II, §6]).

Theorem 7 (Newton polygon, e.g. [2, Ch. II, Prop. 6.3])**.** *Let $f(t) = a_n t^n + a_{n-1}t^{n-1} + \cdots + a_0 \in K[t]$ with $a_n a_0 \neq 0$. The* Newton polygon *of f is the lower convex hull in $\mathbb{R}^2$ of the points $\{(i, v(a_i)) : 0 \le i \le n,\ a_i \neq 0\}$. If the polygon consists of edges of slopes $m_1 < m_2 < \cdots < m_r$ with respective horizontal lengths $\ell_1, \ldots, \ell_r$ ($\sum \ell_j = n$), then f has exactly ℓ_j roots in $\bar{K}$ (counted with multiplicity) whose valuation equals $-m_j$.*

We now apply this to $g(t) = t^3 - \sigma_1 t^2 + \sigma_2 t - \sigma_3$ with $K = \mathbb{Q}(q)$ and $v = \nu_{\mathrm{atom}}$.

Proof of Theorem 2. By Corollary 6, the non-zero eigenvalues of $\Pi_q|_{V_{(4,2)/n=6}}$ are exactly the three roots (in $\overline{\mathbb{Q}(q)}$) of $g(t) = t^3 - \sigma_1 t^2 + \sigma_2 t - \sigma_3$.

We compute the $\mathfrak{a}$-adic valuations of the coefficients of $g(t)$:

- g has a leading 1, so $\nu_{\mathrm{atom}}(\text{coefficient of } t^3) = \nu_{\mathrm{atom}}(1) = 0$.
- $\nu_{\mathrm{atom}}(\text{coefficient of } t^2) = \nu_{\mathrm{atom}}(-\sigma_1) = \nu_{\mathrm{atom}}(\sigma_1) = 0$ by Lemma 3.
- $\nu_{\mathrm{atom}}(\text{coefficient of } t^1) = \nu_{\mathrm{atom}}(\sigma_2) = 1$.
- $\nu_{\mathrm{atom}}(\text{coefficient of } t^0) = \nu_{\mathrm{atom}}(-\sigma_3) = \nu_{\mathrm{atom}}(\sigma_3) = 3$.

The Newton polygon of g is the lower convex hull of the points

$$\big\{(0,3),\ (1,1),\ (2,0),\ (3,0)\big\} \subset \mathbb{R}^2.$$

We claim that all four points lie on the lower convex hull. Indeed, the line through $(0,3)$ and $(2,0)$ has slope $-3/2$ and value $3 - 3/2 = 3/2$ at $i = 1$; the point $(1,1)$ lies strictly below this line, so $(1,1)$ is a vertex of the polygon. All four points are therefore vertices of the lower hull, and the polygon consists of three segments:

$$\begin{array}{lll} (0,3) \to (1,1): & \text{slope } -2, & \text{horizontal length } 1. \\ (1,1) \to (2,0): & \text{slope } -1, & \text{horizontal length } 1. \\ (2,0) \to (3,0): & \text{slope } \ \ 0, & \text{horizontal length } 1. \end{array}$$

By Theorem 7, $g(t)$ has exactly one root of valuation $-(-2) = 2$, exactly one root of valuation $-(-1) = 1$, and exactly one root of valuation 0. These are the three non-zero eigenvalues e_1, e_2, e_3 of $\Pi_q|_{V_{(4,2)/n=6}}$, so (relabelling so that $\nu_{\mathrm{atom}}(e_1) \ge \nu_{\mathrm{atom}}(e_2) \ge \nu_{\mathrm{atom}}(e_3)$):

$$\nu_{\mathrm{atom}}(e_1) = 2, \qquad \nu_{\mathrm{atom}}(e_2) = 1, \qquad \nu_{\mathrm{atom}}(e_3) = 0.$$

This proves the first form of the theorem.

The second form follows by setting $\alpha = e_1/\mathfrak{a}^2$, $\beta = e_2/\mathfrak{a}$, $\gamma = e_3$, all of which lie in $\overline{\mathbb{Q}(q)}$ with $\nu_{\mathrm{atom}} = 0$, and factoring $g(t) = (t - \alpha\mathfrak{a}^2)(t - \beta\mathfrak{a})(t - \gamma)$. $\square$

5 Discussion

5.1 What this gives

This result decodes the cube exponent $\mathfrak{a}(q)^3$ in $\sigma_3(\Pi_q|_{V_{(4,2)/n=6}}) = q^9(1+q)^{21}\mathfrak{a}(q)^3$ as the additive split $3 = 2+1+0$ across three eigenvalue valuations. The cube is not the result of three eigenvalues each carrying a single $\mathfrak{a}$ factor (that would be $1+1+1=3$, also producing a cube but a different distribution); rather, it is the unique distribution forced by the joint atom-valuations of $\sigma_1, \sigma_2, \sigma_3$.

Conceptually, the eigenvalue valuations $(2,1,0)$ form a graded distribution: the "most $\mathfrak{a}$-divisible" eigenvalue carries $\mathfrak{a}^2$, the "middle" carries $\mathfrak{a}$, and the "smallest" carries $\mathfrak{a}^0$. This is structurally distinct from the concentrated distribution $(3,0,0,\ldots,0)$ observed at $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$, where one eigenvalue carries the entire cube and the rest carry no $\mathfrak{a}$ factor.

5.2 What this leaves open

1. **Why graded vs. concentrated?** Theorem 2 pinpoints $(2,1,0)$ at $V_{(4,2)/n=6}$ but explains neither (a) why $V_{(4,3)/n=7}$ and $V_{(5,2)/n=7}$ get the concentrated distribution $(3,0,0,\ldots)$, nor (b) what general principle selects between the two regimes for an arbitrary atom-cube shape. A separate proof cycle on the concentrated case is warranted.

2. **Splitting field of g.** Theorem 2 gives the multiset of valuations of the roots, but does not address whether g splits over $\mathbb{Q}(q)$ or only over a finite extension. The Newton polygon shows the polygon is "regular" (each slope of multiplicity 1), so by general Newton-polygon factorisation theory [2, Ch. II, Cor. 6.5], g factors as a product of three monic linear factors $g = g_1 g_2 g_3$ with $g_j \in \overline{\mathbb{Q}(q)}[t]$ and roots of valuation $2,1,0$ respectively. Whether each g_j is defined over $\mathbb{Q}(q)$ (i.e. each $e_j \in \mathbb{Q}(q)$) is a separate question, equivalent to g splitting over $\mathbb{Q}(q)$.

3. **Connection to the Rosetta Stone.** The factor $\mathfrak{a} = q^2+4q+1$ is the "palindromic atom" of $V_{(3,1)/n=4}$ (the unique non-zero eigenvalue of Π_q on that small-rank Specht module is $q(1+q)^2\mathfrak{a}$, by direct computation). That this atom propagates as $\mathfrak{a}^2$ on one branch and $\mathfrak{a}^0$ on another, under whatever transport carries small-n eigenvalues into large-n eigenvalues, is unexplained. A natural conjecture: the graded distribution $(2,1,0)$ is the "signature of branching" (the three eigenvalues correspond to three layers of $V_{(4,2)/n=6}$ inheriting from successive branchings down to $V_{(3,1)/n=4}$), while the concentrated distribution $(3,0,0,\ldots)$ corresponds to a single "cleanly inherited" eigenvalue.

5.3 Status

This is a theorem in the formal sense: every step is rigorous given (i) the correctness of the symbolic Lagrange interpolation in `verify.py` (exact rational arithmetic, 110 evaluation points exceed the degree bound), (ii) the standard Newton polygon theorem. No appeal to numerics or analogy is made. The conjectural content (concentrated vs. graded; branching-signature interpretation) is clearly separated in §5.2.

References

[1] P. N. Hoefsmit, *Representations of Hecke algebras of finite groups with BN-pairs of classical types*, Ph.D. thesis, University of British Columbia, 1974.

[2] J. Neukirch, *Algebraic Number Theory*, Grundlehren der Mathematischen Wissenschaften 322, Springer-Verlag, 1999.